



A residential microgrid Operational Planning & Sizing

ELEN0445 – Microgrids | Homework 4 Part 1
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16 April 2026

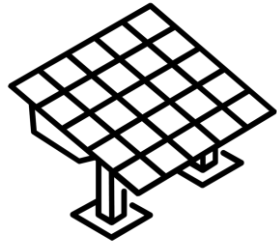
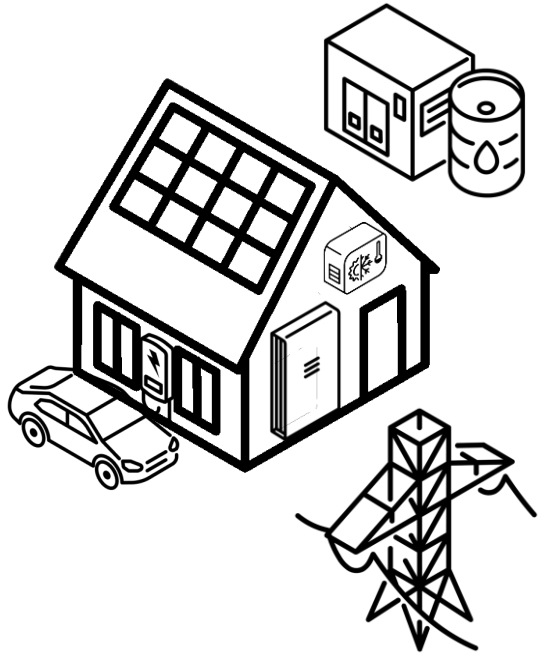


I. Case study

Introduction to the physical problem



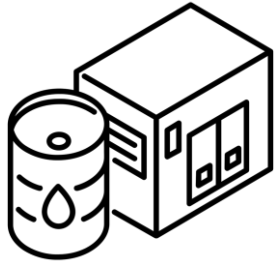
Representation of the house



Photovoltaic panels are the main energy source



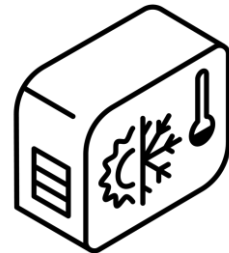
Battery storage system can be used as a slack



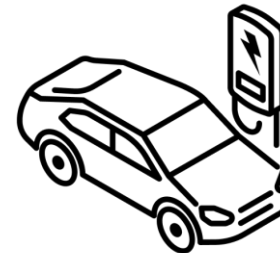
A diesel genset can also provide energy if needed



Load is fixed and should be always satisfied



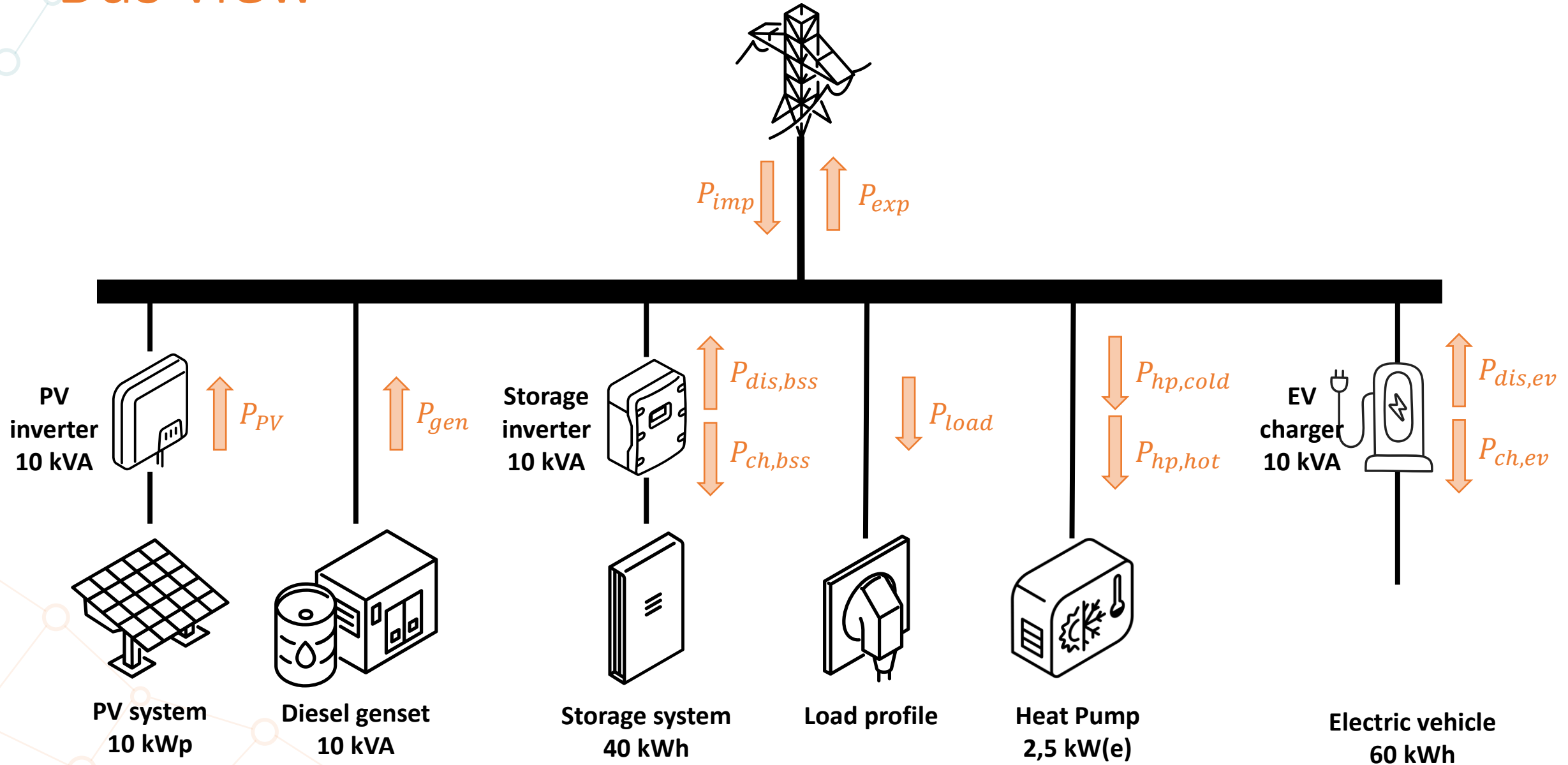
A heat pump is installed for thermal comfort



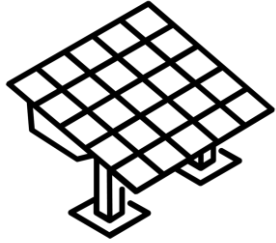
Electric vehicle connects, then should be charged

The system can exchange with the grid at a given price

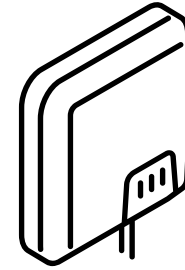
Bus view



Power generation

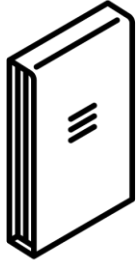


- Main source of energy
- Power profile for MPPs
 - $P_{pv,max}$
- Connected via inverter
- “Free energy”

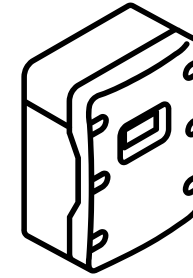


- Inverter connected to the PV panels
- Rated capacity defines the maximum power output
 - $P_{pv,nom}$

Energy storage

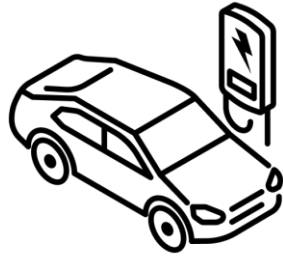


- Should mitigate imbalance
- Fixed capacity
 - C_{bss}
- Constant charging efficiency
 - eff_{bss}
- Bounded SOC's
 - $[SOC_{min\ bss}, SOC_{max\ bss}]$

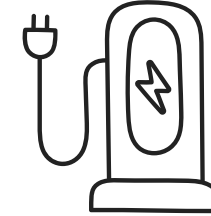


- Inverter connected to the batteries
- Limits the power output
 - $P_{bss\ nom}$

Electric vehicle

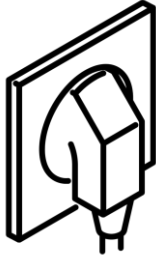


- Not always connected
 - $EV_{connection}$ (binary parameter)
- EV usage is provided with 3 vectors
 - $SOC_{ev,i}$ t_{arr} , t_{dep}
- Should be charged when leaving
 - $SOC_{ev\ target}$
- Fixed capacity and efficiency
 - C_{ev} , eff_{ev}
- Bounded SOC's
 - $[SOC_{min\ ev}, SOC_{max\ ev}]$

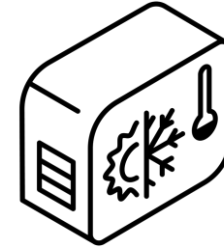


- Charger connected to the electric vehicle
- Limited charging power
 - $P_{ev\ nom}$

Fixed load and Heat pump

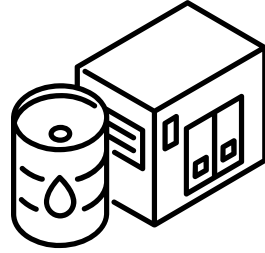


- Represents plugged-in appliances
- Defined by a profile
 - P_{load}
- Scaled by your yearly consumption
- Should always be satisfied



- Should manage temperature
- Power limited by device
 - $P_{hp\ max}$
- Home thermal capacity [kWh/°C]
 - C_{hp}
- Coefficient of performance
 - COP_{hp}
- Bounded temperature
 - $[T_{set} - \Delta T_{max}, T_{set} + \Delta T_{max}]$

Controllable generation



- Contingency usage
- Maximum/minimum power
 - $P_{max,gen}$
- Fuel costs when used
- Avoid repetitive on-off switches
- Power output should maximize efficiency



- Represents plugged-in appliances
- Defined by a profile
 - P_{load}
- Scaled by your yearly consumption
- Should always be satisfied



Objectives of the controller

This is what a good controller should achieve, in decreasing order of priority

Ensure feasibility: Power/energy bounds, demand satisfaction,...

Approach optimality: final cost, asset management, genset operation,...

Improve reliability: robustness, more realistic model of appliances,...



II. Operational Planning

Description of the first part of the homework

Objective

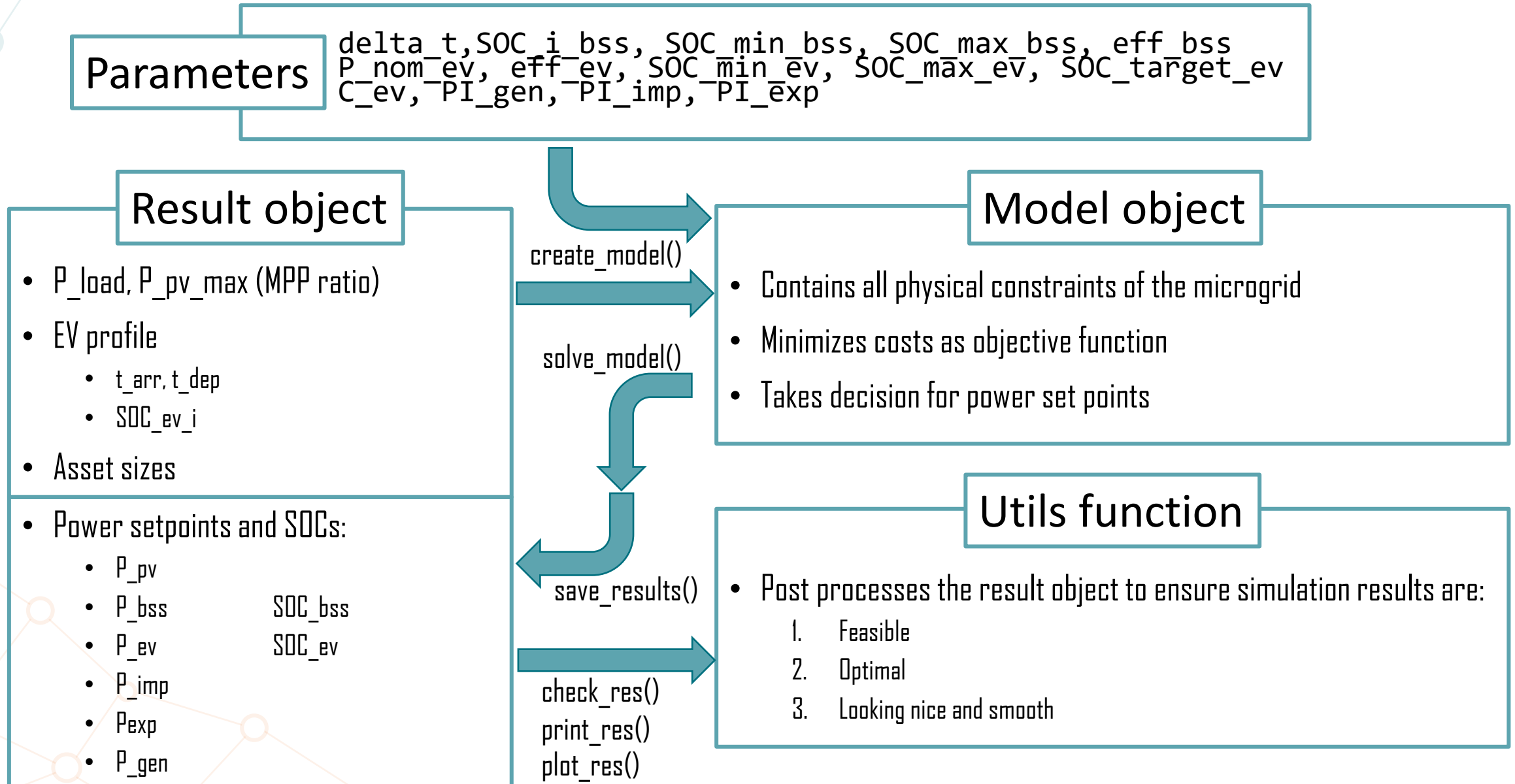


Using perfect forecast and given equipment sizes, define the complete optimal power profile of all assets at once.

➔ In practice, this step would be done daily, this is called day-ahead operational planning.

➔ In this part, we will consider all available data at once to get yearly costs directly.

Code diagram





How to access data in the Results object

Inputs

- `res.P_load[t]` = Load power at time t
- `res.P_pv_max[t]` = Max available PV power at time t
- `res.P_loss[t]` = Power lost through house envelope
- `res.T_set[t]` = House temperature setpoint
- `res.EV_connected[t]` = EV connected at time t
- `res.t_arr[t]` = 1 if EV connects at time t
- `res.t_dep[t]` = 1 if EV leaves at time t
- `res.SOC_i_ev[t]` = Arrival SOC of the EV

Outputs

- `res.P_pv[t]` : Power of the PV system [kW]
- `res.P_gen[t]` : Power of the generator [kW]
- `res.P_bss[t]` : Power of the battery [kW]
- `res.P_ev[t]`: Power of the EV [kW]
- `res.P_imp[t]` : Power of the battery [kW]
- `res.P_exp[t]`: Power of the EV [kW]
- `res.P_hp_hot[t]`: Power for heating with HP [kW]
- `res.P_hp_cold[t]`: Power for cooling with HP [kW]
- `res.T_hp[t]`: House temperature [°C]
- `res.SOC_bss[t]`: Energy in the battery [kWh]
- `res.SOC_ev[t]`: Energy in the EV [kW]
- `res.objective`: Value of the objective function



Pay attention to solving time

- The decision horizon will go until 1 year, so the running time will grow drastically
- You should keep a linear formulation for the solution to be tractable
- Avoid non-convexity at all costs:
 - ~~➤ $P_{charge} P_{discharge} = 0$~~
- Use of binary variables is also unwanted if you can avoid them



Guidelines for operational planning

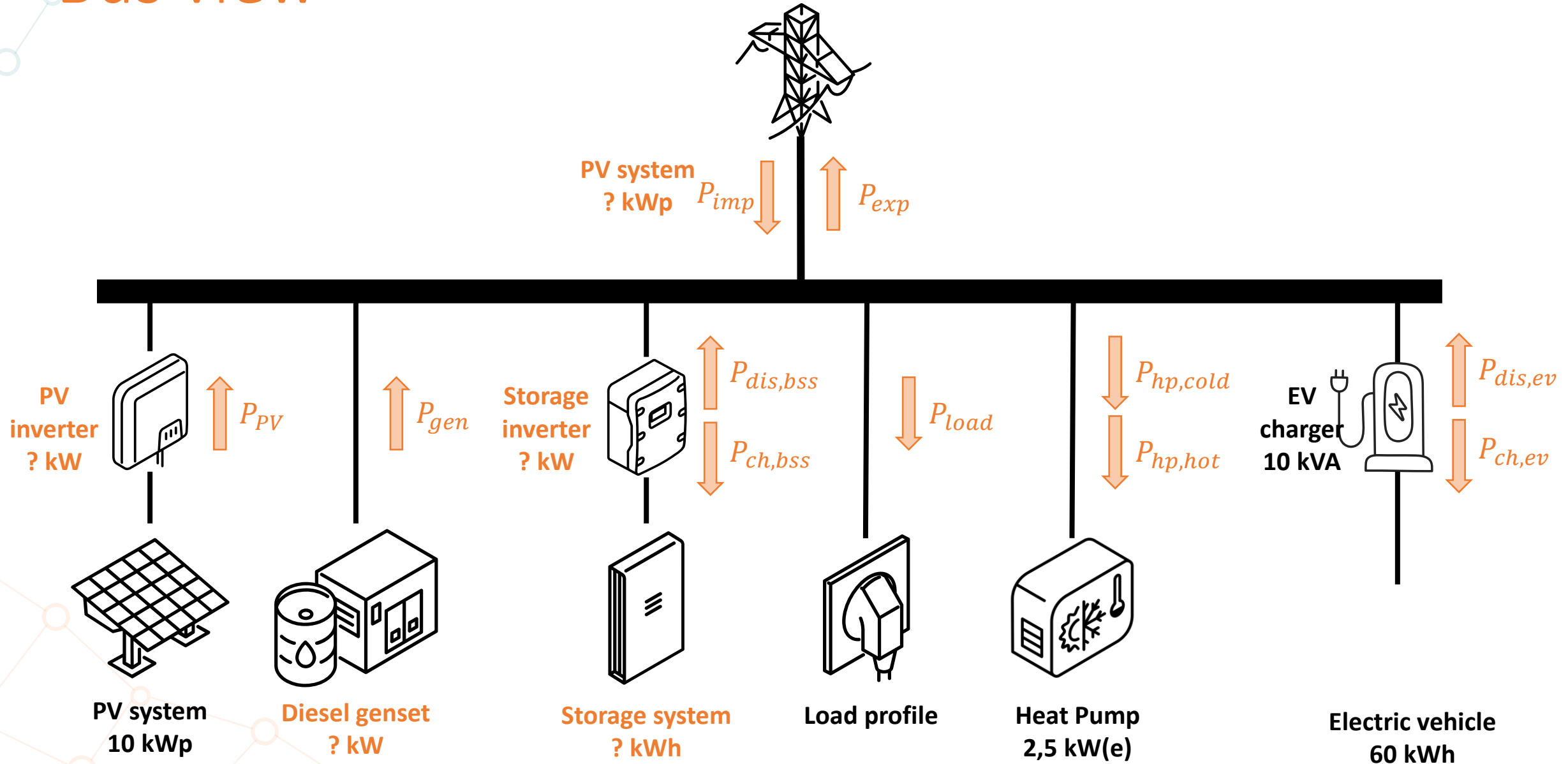
0. Define the yearly load, EV_usage, and comfort temperature range
1. Solve the operational planning problem using a linear programming formulation. Optimize the use of each device to minimize OPEX.
2. Compute the yearly system costs and present some results.
3. Show how evolving energy prices would affect the results with this given system. Decorrelate import and export prices.



III. Sizing

Description of the second part of the homework

Bus view





Main changes from the last part

- **Asset sizes are now variables** and not parameters
- Parameters of the model now include investment horizon and investment prices for each technology
- CAPEX should be included in the objective function
- If relevant, you should also add constraints and variables to the model



Guidelines for sizing – Part 1

1. Reformulate the optimization problem of the previous section as a sizing problem. Additional constraints could be required, and the objective function must be adapted to include scaled CAPEX.
2. Compute the optimal sizing of your system, this is your base case.
3. Compare this first result with an isolated microgrid and a case where exports are not paid
4. On the base case, show how a varying CAPEX budget impacts the investment



Guidelines for sizing – Part 2

5. On the base case, make the load vary (Base, HP and EV).
6. Implement the effect of varying energy costs during the investment horizon.
7. Bonus: model CO₂ emissions and consider them in your optimization.



IV. Forecasting

Description of the third part of the homework
This will be presented on May 7th



V. Report

Report guidelines



1. **Maximum 9 pages**
2. Pay attention to the graphs and results you present
3. Be concise, focus on your decisions and results, not on the case study
 - Try to present the result in the most adequate possible way
 - Use graphs to show dependencies
 - Use tables to summarize and compare results
 - If a proposed lead is less interesting, do not present, but justify
4. Deadline: 23:59, 3 days before the exam